Experiment 1
Optical free-beam transmission system

Table of contents

1. introduction 2
2 The pn transition 3
3 The optical transmission system 7
4. radiation properties 8
5. experimental setup 9
6. practical experiments 10
7 Organizational notes 11
8. literature 11

1. Introduction

Optical communication using optoelectronic components has been widely used in the field of information technology since the development of lasers and low-attenuation optical fibers. The aim of this practical experiment is to investigate the functionality and basic physical properties of some optoelectronic components and the optical signal transmission systems built with them.

An optical transmission system basically consists of three elements: An electro-optical converter, which converts an electrical signal into an optical signal, an optical transmission path and an opto-electrical converter, which converts the optical signal back into an electrical signal (Fig. 1). Laser diodes and luminescence diodes are used as converters on one side and photodiodes on the other. Dielectric waveguides in the form of glass fibers are usually used as the optical transmission medium. The attenuation of these glass fibers can reach values of less than 0.2 dB/km at a wavelength of λ = 1.3 µm.

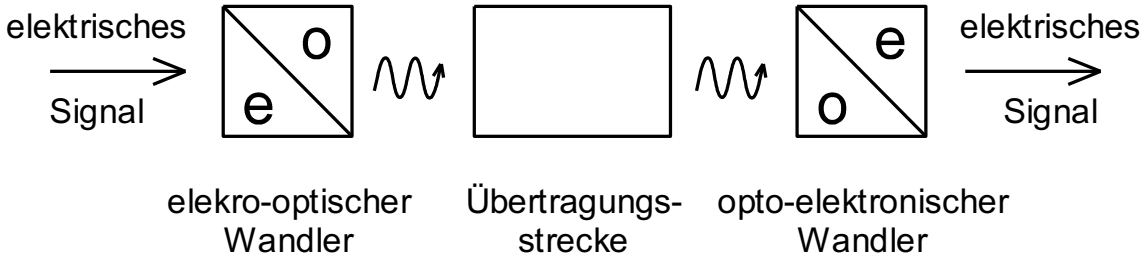


Fig. 1: Principle of an optical transmission system

In this practical experiment, the properties of typical optoelectronic components will be discussed. For this purpose, the characteristics of a photodiode and a luminescent diode (LED) are determined and evaluated experimentally. In particular, the transmission properties of the elements and those of an optical transmission system constructed with them will be examined in more detail.

The questions listed below serve to familiarize students with the topic of this experiment and should be discussed during the practical course. In preparation for the experiment, the basic electrical and optical properties of the following components should first be discussed:

- pn diode
- Photodiode
- Luminescent diode
- Laser diode
- Fiber optic cable

2. The pn transition

The electronic properties of a pn-contact are of central importance for the components discussed below. First, the band diagram of a pn junction is shown and the characteristic equation is given. Then the optoelectronic properties of a pn-contact are summarized for radiation reception and radiation emission.

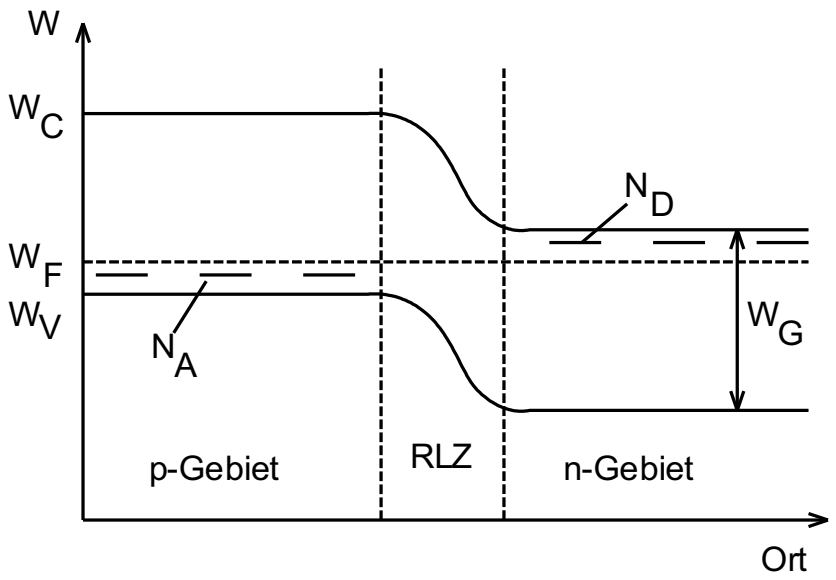


Fig. 2: Band diagram of a pn junction

If an n-doped semiconductor is brought into contact with a p-doped semiconductor, the Fermi levels (W_F) equalize in thermodynamic equilibrium. This results in the band diagram shown in Fig. 2. Between the n-doped orbital region with electrons as majority carriers and the p-doped orbital region with holes as majority carriers, a space charge zone is formed that is depleted of free charge carriers. Fig. 3 shows the current-voltage (I-V) characteristic of a pn junction, which is described by the equation:

$$I = I_s\left(e^{U/U_T} - 1\right)$$

is described.

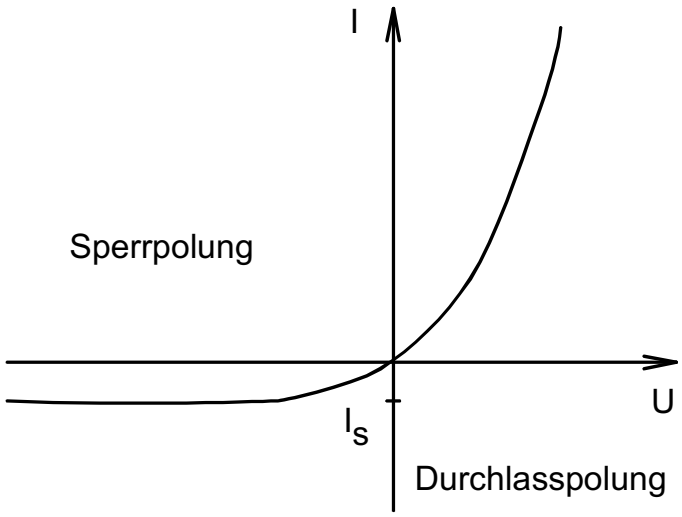


Fig. 3: Characteristic curve of the pn junction

These are $U_T = \frac{k\,T}{e}$ and $I_S = eA\left(\frac{n_i^2\,L_p}{N_D\,\tau_p} + \frac{n_i^2\,L_n}{N_A\,\tau_n}\right)$,

with $L_p = \sqrt{D_p \tau_p}$ and $L_n = \sqrt{D_n \tau_n}$

A: Cross-sectional area of the diode

I_S: Saturation current

n_i: Intrinsic line concentration

N_D , N_A: Doping concentration

τ_p , τ_n: Charge carrier life

D_p , D_n: Diffusion coefficients

e Elementary charge

The physical basis of the optoelectronic components considered in this experiment is the interaction of optical radiation with semiconductor materials. Photon energies $W = h\nu$ in the order of the band gap energy W_g , i.e. in the order of about 1 eV (with Planck's quantum of action *h* and frequency of the optical radiation ν), are therefore of particular importance. This corresponds to a wavelength of approximately λ = 1.24 µm, whereby the conversion between the radiation energy and the wavelength of the photons is a good approximation:

$$\frac{W_g}{\mathrm{eV}} = \frac{1{,}24}{\frac{\lambda_g}{\mu\mathrm{m}}} \text{ applies.}$$

The steep increase in the absorption of optical radiation for the wavelength range $\lambda < \lambda_g$ is characteristic. The basic physical elementary process is the generation of a free electron-hole pair for energies $W \geq W_g$ (see Fig. 4 (a)).

The reverse process to the generation of free charge carriers is the recombination of free electrons in the conduction band with free states in the valence band, the so-called holes. The recombination can take place radiatively, i.e. lead to the emission of a photon with an energy that is approximately that of the band gap. A distinction is made between spontaneous emission (see Fig. 4 (b)), which takes place without the involvement of external radiation, and stimulated emission. In stimulated emission, recombination processes are stimulated by photons that are already present. The newly emitted photons have the same energy and phase position as the stimulating photons (see Fig. 4 (c)). In general, emission can be seen as the inverse process to absorption, formally leading to a negative absorption coefficient α .

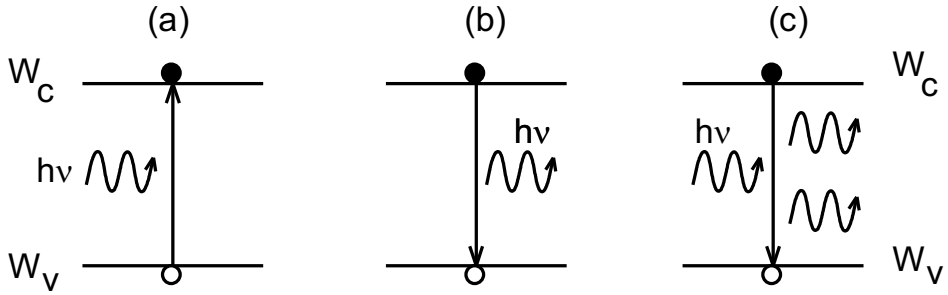


Fig. 4: (a) Absorption, (b) spontaneous and (c) induced emission in the band diagram

The three elementary processes discussed form the basis of the following optoelectronic components:

- Photodetectors (Fig. 4(a))
- Luminescent diodes (Fig. 4(b))
- Laser diodes (Fig. 4 (c))

These components are used as converters between the electrical and optical signal levels.

In photodiodes (Fig. 5), the internal electric field of the space charge zone is used to separate the optically generated charge carriers. A photocurrent is generated as an impressed current that flows even without an external voltage. Conversely, the imprinting of a current - as in the case of the LED - can lead to the emission of photons that are generated in the pn junction due to radiative recombination.

In technical components, the pn contact is often replaced by an intrinsic transition zone in order to achieve higher efficiency. Such a component is referred to as a pin photodiode. In Schottky

photodiodes, on the other hand, the junction of a metal-semiconductor contact is used to generate a photocurrent.

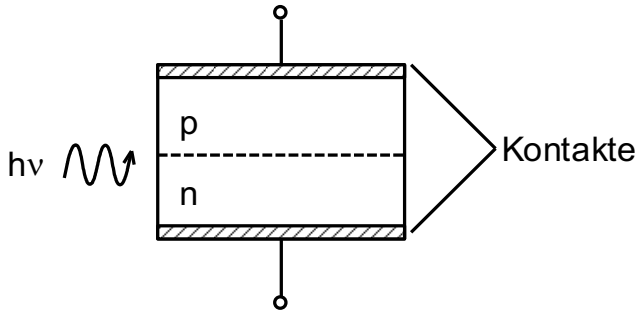


Fig. 5: pn photodiode

Question 1: Which properties of the pn junction result from the band diagram and the characteristic curves?

Question 2: How does the characteristic curve shown in Fig. 3 change when the photodiode is illuminated?

Question 3: When is there a linear relationship between the optical power P and the electrical measured variable I?

Question 4: How can the transmission properties of the components (LED, photodiode) be determined?

3. The optical transmission system

An optical transmission system can be constructed from the components described above. This consists of an LED, transmission path and photodiode (PD). Fig. 6 outlines the basic principle of the system and shows the typical transmission behavior.

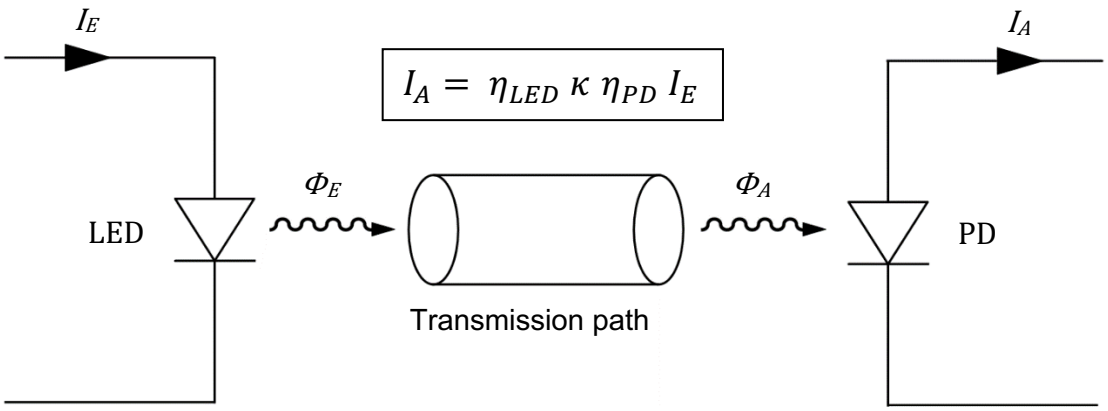


Fig. 6: Practical realization of the optical transmission system

If an input current I_E flows through the radiation emitter, the outgoing photon flux according to a conversion efficiency η_{LED} outgoing photon flux Φ_E passes through the transmission path with the transmission efficiency κ. The flux $\kappa\, \Phi_E = \Phi_A$ reaches the photodetector and generates the electrical output current I with the conversion efficiency η_{LED} the electrical output current I_A .

Question 1: How is the transmission characteristic of the overall system determined from the individual components?

Question 2: When is the system a linear, time-invariant system and how can this be verified experimentally?

Question 3: What changes if the length of the transmission path is changed?

4. Radiation properties

The radiation intensity I_e and the light intensity I_v depend, among other things, on the emission angle ε_1 , which results from the shape of the LED. Different LED designs therefore provide different radiation properties. The progression of the luminous intensity as a function of the angle ε_1 is shown in polar coordinates in Fig. 7. The LED with the luminous intensity curve shown in Fig. 7(a) has a colored, diffusely scattering plastic housing and approximately fulfills the characteristics of a Lambert emitter. The beam angle is $\varphi = 60°$.

$$I_\upsilon(\varepsilon_1) = I_\upsilon(0)\cos\varepsilon_1$$

The beam angle or opening angle □□kennzeichnet is the angle at which the light intensity drops to half of the maximum value. The LED with the light intensity curve shown in Fig. 7(b) has a highly transparent, uncolored housing and emits with a beam angle of only φ = 12°.

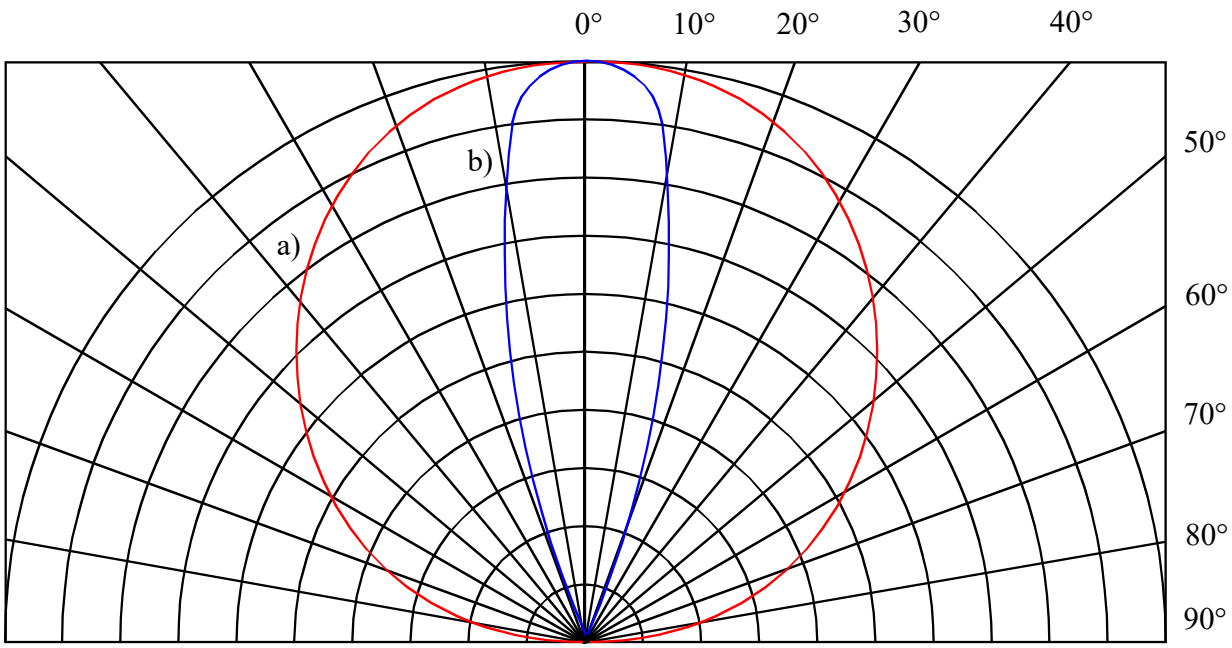


Fig. 7: Radiation properties of two LEDs a)φ = 60°, b)φ = 12°

Question 1:	Where can both LEDs be used?
Question 2:	What designs do they have?

5. Experimental setup

This chapter describes the most important measuring instruments used in this experiment. Fig.8 (left) shows the experimental setup. This consists of a light-emitting diode and a photodiode whose distance *r* from each other can be variably adjusted via a rail. The two Hewlett-Packard type 3478A multimeters are used to measure current and voltage. On the right-hand side of the Fig.8 you can see the voltage source with which the light-emitting diode is operated.

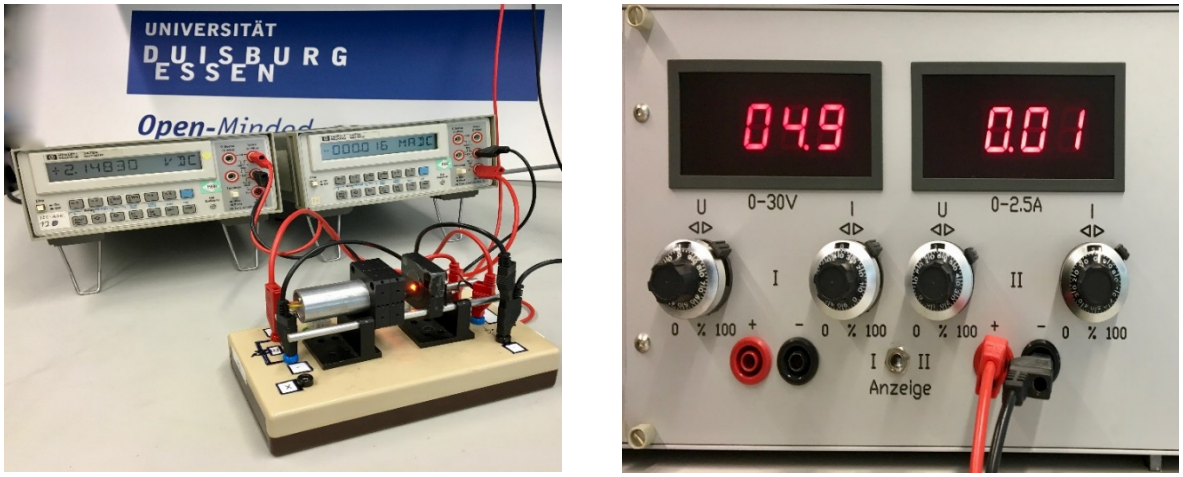


Fig. 8: Optical free-beam transmission system and voltage source

An oscilloscope with function generator is used to transmit frequency-dependent signals with the aid of the LED and to measure these frequency-dependent variables. In addition to the amplitude and the offset, the frequency and the signal curve can be set. You can choose between sinusoidal, triangular and rectangular waveforms. The Keysight EDUX1022G oscilloscope is used in this practical course (Fig.9).



Fig. 9: Keysight EDUX1022G oscilloscope with function generator

6. Practical experiments

The following practical experiments are to be carried out under supervision during the test date:

a) LED, *I-V* characteristic: The current-voltage characteristic of the LED must be determined. Use a digital multimeter (Hewlett Packard) to determine the current through the diode and a second digital multimeter (Kontron) to measure the voltage across the LED.

Apply: I(U)

Determine: U_D

b) PD, *I-P*$_{in}$: The $I(P_{PD})$ characteristic curve of the PD is to be recorded via the proportionality of the distance r between the LED and PD and the power P_{PD} generated at the photodiode. The load resistance at the PD is R = 20 k□. To do this, the voltage across the load resistor $R = U / I_{PD}$ must be recorded with the HP digital multimeter and the distance r must be changed appropriately. The voltage at the power supply unit for the LED should be set to U_{Netz} = 7 V.

Apply: I(r), I_{PD} (P_{PD}), (P_{PD} in (W) and (dBm))

c) Overall efficiency: For the smallest adjustable distance r_{min} , the electrical power consumed by the LED P_{LED} (U_{Netz} = 7 V, I_{LED} = ?) and the electrical power emitted by the photodiode P_{PD} (R = 20 k□, I_{PD} = ?) should be determined.

Determine: P_{LED} , P_{PD} (r_{min}), □□$(r)_{min}$

d) Rise and fall time: For these AC measurements, the signal generator (offset 5 V, amplitude 5 V, square-wave voltage) must be connected to the LED via the series resistor. The dual-channel oscilloscope should tap the voltage across the LED and series resistor on channel 1 and the voltage across the resistor on the PD on channel 2. Triggering takes place on channel 1. At a frequency of ν = 20 kHz, the rise time t_{rise} and the fall time t_{fall} are to be determined. Provided that the system can be described by an RC *low-pass filter*, it is possible to determine the cut-off frequency f_{gr} of the system using the following equation:

$$f_{gr} = \frac{\ln 0{,}9 - \ln 0{,}1}{2\pi \cdot t_{rise/fall}} \cong \frac{0{,}35}{\frac{t_{rise} + t_{fall}}{2}}$$

Determine: *Rise time t_{rise}*
Fall time t_{fall}
Cut-off frequency f_{gr}

e) Signal delay, phase shift: The frequency range ν = 1 kHz ... 80 kHz should be scanned with the signal generator (offset 5 V, amplitude 5 V, sinusoidal voltage) and the signal delay t_0 between channel 1 and channel 2 should be measured.

Apply: t_0 □□□, $\phi(\nu)$

f) Determining the cut-off frequency: The frequency must be increased at the signal generator (offset 5 V, amplitude 5 V, sinusoidal voltage) from ν = 250 Hz until the amplitude of the signal at the PD has fallen by 3 dB.

Determine: Cut-off frequency: f_{3dB}

g) Optical transmission system: Construct an optical free-beam transmission system with an LED as the signal source and a photodiode as the signal receiver and then transmit audio signals. The length of the free-beam path should be at least 1m.

7. Organizational notes

- Participants must bring a calculator, pencil, 30-cm ruler and linearly divided graph paper to the practical course.
- The answers to the comprehension questions within the internship printout should be discussed by the participants as part of the preparation.
- If all copies of the specified literature are on loan in the library, it is also possible to prepare an experiment using the reference copies.
- The tasks listed above are to be completed during the practical course. The written elaboration (graphs, explanations) is to be postponed until the end of the practical course.

8. Literature

[1] H.-G. Unger, W. Schulz, G. Weinhausen, *Elektronische Bauelemente und Netzwerke I,* Vieweg & Sohn, Braunschweig 1979.

[2] R. Paul, *Optoelektronische Halbleiterbauelemente,* Teubner Studienskripte, Stuttgart 1985.

[3] K.J. Ebeling, *Integrated Optoelectronics,* Springer Verlag, Berlin 1989.

[4] M. Bleicher, *Halbleiter-Optoelektronik,* Dr. Alfred Hüthig Verlag, Heidelberg 1986.

[5] H. P. Lüke, *Signal Transmission,* Springer Verlag, Berlin 1988.

[6] Donges and R. Noll, *Lasermeßtechnik,* Dr. Alfred Hüthig Verlag, Heidelberg 1993.

[7] Frank Thuselt, *Physics of semiconductor devices,* Springer
https://link.springer.com/book/10.1007/978-3-642-20032-8